

EXP. NO: 5

EDA – DATA VISUALIZATION

AIM:

To visualize the dataset using different plots to identify patterns, distributions, relationships, and anomalies.

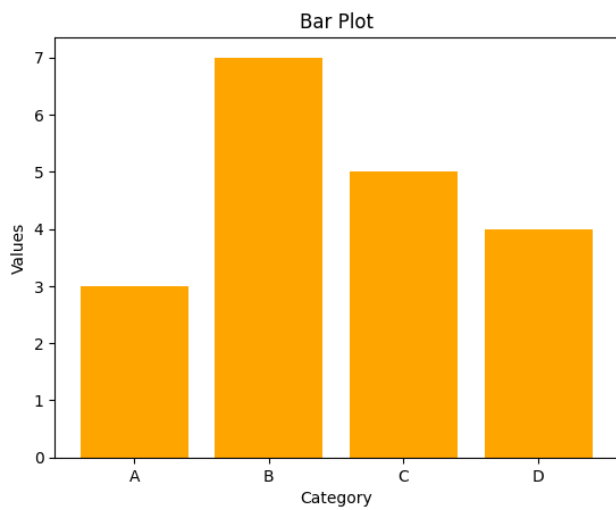
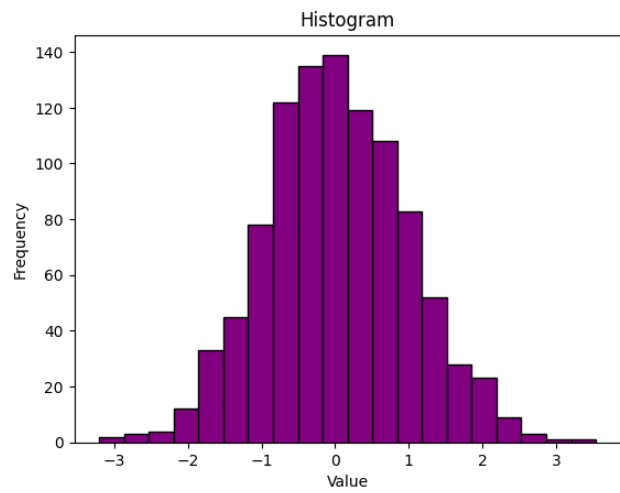
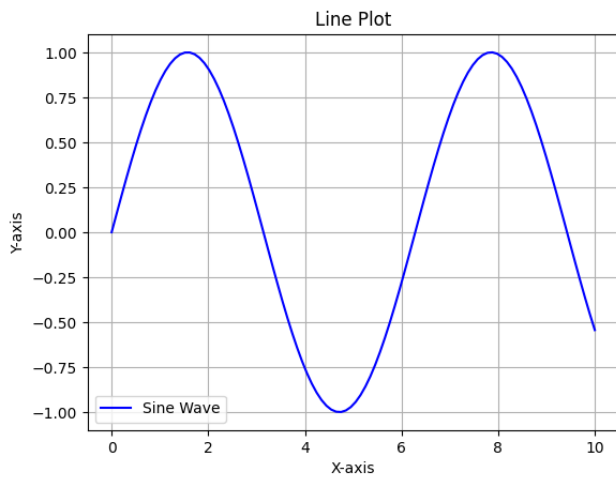
PROCEDURE:

1. Import Pandas, NumPy, and Matplotlib libraries.
2. Load the dataset using `pd.read_csv()`.
3. Display the first few records using `head()`.
4. Generate a Line Chart to show trends in data.
5. Generate a Bar Chart to compare different categories.
6. Generate a Histogram to show the distribution of numerical data.
7. Add suitable labels, titles, and legends to the plots.
8. Display the visualizations using `plt.show()`.

PROGRAM:

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
df = pd.read_csv("/content/data.csv")
print(df.head())
# LINE CHART
x = np.linspace(0, 10, 100)
y = np.sin(x)
plt.plot(x, y, label="Sine Wave")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Line Plot")
plt.legend()
plt.show()
# BAR CHART
categories = ['A', 'B', 'C', 'D']
values = [3, 7, 5, 4]
plt.bar(categories, values)
plt.xlabel("Category")
plt.ylabel("Values")
plt.title("Bar Plot")
plt.show()
# HISTOGRAM
data = np.random.randn(1000)
plt.hist(data, bins=20, edgecolor='black')
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.title("Histogram")
plt.show()
```

OUTPUT:



RESULT:

Thus, the given program was successfully executed and different visualizations were generated to understand patterns and distributions in the data.